HEXADIGITALL TECHNOLOGIES

OFFICIAL CURRICULUM GUIDE

Advanced Ansible Automation & IaC

Master Infrastructure as Code (IaC) with Ansible. Automate configuration management and deployment.

DURATION **20 Weeks**	LEVEL **Advanced**	STUDY TIME **6 hours/week + labs**
SCHOOL **Hexadigitall Academy**	PREPARED ON **09 April 2026**	SOURCE **Advanced Ansible Automation & IaC**

PROGRAM SNAPSHOT

Advanced Ansible Automation & IaC

Master Infrastructure as Code (IaC) with Ansible. Automate configuration management and deployment.

DURATION
20 Weeks

LEVEL
Advanced

STUDY TIME
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SCHOOL
Hexadigitall Academy

Welcome to Advanced Ansible Automation & IaC!

This curriculum is designed to take you from core understanding to confident delivery through weekly applied practice, measurable outcomes, and portfolio evidence.

Each week builds progressively with practical tasks, implementation checkpoints, and reflection points so you can convert knowledge into repeatable professional performance.

Your success is our priority. Stay consistent with weekly execution, document your work, and use feedback loops to continuously improve your delivery quality.

Essential Resources

- CI/CD pipeline patterns and deployment strategy reference implementations
- Infrastructure automation templates and reliability engineering guides
- Observability stack setup instructions and incident response playbooks

Learning Roadmap

- Early Weeks: Version control workflows, CI foundation, and automation basics
- Middle Weeks: Advanced CI/CD, container orchestration, and pipeline optimization
- Late Weeks: Production reliability, observability, and operational excellence
- Roles, Collections, and Reusable Automation Design
- Secrets Management with Ansible Vault
- Package, Service, and File Automation at Scale
- Inventories, Credentials, RBAC, and Workflow Templates
- Dynamic Inventory for Cloud and Virtual Environments
- CI/CD Integration for Ansible Projects
- Network Automation with Vendor Modules and APIs
- Windows Automation with WinRM and PowerShell Patterns
- Container and Kubernetes Automation Use Cases

Detailed Weekly Curriculum

WEEK 1

6 hours/week + labs

Ansible Control Node Setup, Inventory Structure, and SSH Access

LEARNING OUTCOMES

- Analyze the principles of Ansible Control Node Setup, Inventory Structure, and SSH Access and link them to course outcomes at advanced depth with architecture-level decision quality.
- Evaluate Ansible Control Node Setup, Inventory Structure, and SSH Access in a guided scenario using realistic tools, constraints, and quality gates.
- Design trade-offs, risks, and decision points for Ansible Control Node Setup, Inventory Structure, and SSH Access, then record rationale for stakeholder review.
- Justify a portfolio-ready pipeline reliability report for Ansible Control Node Setup, Inventory Structure, and SSH Access with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 2

6 hours/week + labs

Host Patterns, Group Variables, and Fact Gathering

LEARNING OUTCOMES

- Analyze the principles of Host Patterns, Group Variables, and Fact Gathering and link them to course outcomes at advanced depth with architecture-level decision quality.
- Evaluate Host Patterns, Group Variables, and Fact Gathering in a guided scenario using realistic tools, constraints, and quality gates.
- Design trade-offs, risks, and decision points for Host Patterns, Group Variables, and Fact Gathering, then record rationale for stakeholder review.
- Justify a portfolio-ready pipeline reliability report for Host Patterns, Group Variables, and Fact Gathering with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 3

6 hours/week + labs

Playbook Structure, Tasks, and Idempotent Execution

LEARNING OUTCOMES

- Analyze the principles of Playbook Structure, Tasks, and Idempotent Execution and link them to course outcomes at advanced depth with architecture-level decision quality.
- Evaluate Playbook Structure, Tasks, and Idempotent Execution in a guided scenario using realistic tools, constraints, and quality gates.
- Design trade-offs, risks, and decision points for Playbook Structure, Tasks, and Idempotent Execution, then record rationale for stakeholder review.
- Justify a portfolio-ready pipeline reliability report for Playbook Structure, Tasks, and Idempotent Execution with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 4

6 hours/week + labs

Variables, Templates, Loops, and Conditionals

LEARNING OUTCOMES

- Analyze the principles of Variables, Templates, Loops, and Conditionals and link them to course outcomes at advanced depth with architecture-level decision quality.
- Evaluate Variables, Templates, Loops, and Conditionals in a guided scenario using realistic tools, constraints, and quality gates.
- Design trade-offs, risks, and decision points for Variables, Templates, Loops, and Conditionals, then record rationale for stakeholder review.
- Justify a portfolio-ready pipeline reliability report for Variables, Templates, Loops, and Conditionals with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 5

6 hours/week + labs

Handlers, Tags, Blocks, and Error Recovery

LEARNING OUTCOMES

- Analyze the principles of Handlers, Tags, Blocks, and Error Recovery and link them to course outcomes at advanced depth with architecture-level decision quality.
- Evaluate Handlers, Tags, Blocks, and Error Recovery in a guided scenario using realistic tools, constraints, and quality gates.
- Design trade-offs, risks, and decision points for Handlers, Tags, Blocks, and Error Recovery, then record rationale for stakeholder review.
- Justify a portfolio-ready pipeline reliability report for Handlers, Tags, Blocks, and Error Recovery with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 6

6 hours/week + labs

Roles, Collections, and Reusable Automation Design

LEARNING OUTCOMES

- Analyze the principles of Roles, Collections, and Reusable Automation Design and link them to course outcomes at advanced depth with architecture-level decision quality.
- Evaluate Roles, Collections, and Reusable Automation Design in a guided scenario using realistic tools, constraints, and quality gates.
- Design trade-offs, risks, and decision points for Roles, Collections, and Reusable Automation Design, then record rationale for stakeholder review.
- Justify a portfolio-ready pipeline reliability report for Roles, Collections, and Reusable Automation Design with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 7

6 hours/week + labs

Secrets Management with Ansible Vault

LEARNING OUTCOMES

- Analyze the principles of Secrets Management with Ansible Vault and link them to course outcomes at advanced depth with architecture-level decision quality.
- Evaluate Secrets Management with Ansible Vault in a guided scenario using realistic tools, constraints, and quality gates.
- Design trade-offs, risks, and decision points for Secrets Management with Ansible Vault, then record rationale for stakeholder review.
- Justify a portfolio-ready pipeline reliability report for Secrets Management with Ansible Vault with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 8

6 hours/week + labs

Package, Service, and File Automation at Scale

LEARNING OUTCOMES

- Evaluate the principles of Package, Service, and File Automation at Scale and link them to course outcomes at advanced depth with architecture-level decision quality.
- Design Package, Service, and File Automation at Scale in a guided scenario using realistic tools, constraints, and quality gates.
- Optimize trade-offs, risks, and decision points for Package, Service, and File Automation at Scale, then record rationale for stakeholder review.
- Justify a portfolio-ready pipeline reliability report for Package, Service, and File Automation at Scale with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 9

6 hours/week + labs

User Provisioning, Access Baselines, and Server Hardening

LEARNING OUTCOMES

- Evaluate the principles of User Provisioning, Access Baselines, and Server Hardening and link them to course outcomes at advanced depth with architecture-level decision quality.
- Design User Provisioning, Access Baselines, and Server Hardening in a guided scenario using realistic tools, constraints, and quality gates.
- Optimize trade-offs, risks, and decision points for User Provisioning, Access Baselines, and Server Hardening, then record rationale for stakeholder review.
- Justify a portfolio-ready pipeline reliability report for User Provisioning, Access Baselines, and Server Hardening with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 10

6 hours/week + labs

AWX or Automation Controller Architecture and Job Templates

LEARNING OUTCOMES

- Evaluate the principles of AWX or Automation Controller Architecture and Job Templates and link them to course outcomes at advanced depth with architecture-level decision quality.
- Design AWX or Automation Controller Architecture and Job Templates in a guided scenario using realistic tools, constraints, and quality gates.
- Optimize trade-offs, risks, and decision points for AWX or Automation Controller Architecture and Job Templates, then record rationale for stakeholder review.
- Justify a portfolio-ready pipeline reliability report for AWX or Automation Controller Architecture and Job Templates with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 11

6 hours/week + labs

Inventories, Credentials, RBAC, and Workflow Templates

LEARNING OUTCOMES

- Evaluate the principles of Inventories, Credentials, RBAC, and Workflow Templates and link them to course outcomes at advanced depth with architecture-level decision quality.
- Design Inventories, Credentials, RBAC, and Workflow Templates in a guided scenario using realistic tools, constraints, and quality gates.
- Optimize trade-offs, risks, and decision points for Inventories, Credentials, RBAC, and Workflow Templates, then record rationale for stakeholder review.
- Justify a portfolio-ready pipeline reliability report for Inventories, Credentials, RBAC, and Workflow Templates with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 12

6 hours/week + labs

Dynamic Inventory for Cloud and Virtual Environments

LEARNING OUTCOMES

- Evaluate the principles of Dynamic Inventory for Cloud and Virtual Environments and link them to course outcomes at advanced depth with architecture-level decision quality.
- Design Dynamic Inventory for Cloud and Virtual Environments in a guided scenario using realistic tools, constraints, and quality gates.
- Optimize trade-offs, risks, and decision points for Dynamic Inventory for Cloud and Virtual Environments, then record rationale for stakeholder review.
- Justify a portfolio-ready pipeline reliability report for Dynamic Inventory for Cloud and Virtual Environments with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 13

6 hours/week + labs

CI/CD Integration for Ansible Projects

LEARNING OUTCOMES

- Evaluate the principles of CI/CD Integration for Ansible Projects and link them to course outcomes at advanced depth with architecture-level decision quality.
- Design CI/CD Integration for Ansible Projects in a guided scenario using realistic tools, constraints, and quality gates.
- Optimize trade-offs, risks, and decision points for CI/CD Integration for Ansible Projects, then record rationale for stakeholder review.
- Justify a portfolio-ready pipeline reliability report for CI/CD Integration for Ansible Projects with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 14

6 hours/week + labs

Testing Playbooks with Molecule, Linting, and Validation

LEARNING OUTCOMES

- Design the principles of Testing Playbooks with Molecule, Linting, and Validation and link them to course outcomes at advanced depth with architecture-level decision quality.
- Optimize Testing Playbooks with Molecule, Linting, and Validation in a guided scenario using realistic tools, constraints, and quality gates.
- Architect trade-offs, risks, and decision points for Testing Playbooks with Molecule, Linting, and Validation, then record rationale for stakeholder review.
- Defend a portfolio-ready pipeline reliability report for Testing Playbooks with Molecule, Linting, and Validation with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 15

6 hours/week + labs

Patch Management, Compliance Checks, and Drift Control

LEARNING OUTCOMES

- Design the principles of Patch Management, Compliance Checks, and Drift Control and link them to course outcomes at advanced depth with architecture-level decision quality.
- Optimize Patch Management, Compliance Checks, and Drift Control in a guided scenario using realistic tools, constraints, and quality gates.
- Architect trade-offs, risks, and decision points for Patch Management, Compliance Checks, and Drift Control, then record rationale for stakeholder review.
- Defend a portfolio-ready pipeline reliability report for Patch Management, Compliance Checks, and Drift Control with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
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- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 16

6 hours/week + labs

Network Automation with Vendor Modules and APIs

LEARNING OUTCOMES

- Design the principles of Network Automation with Vendor Modules and APIs and link them to course outcomes at advanced depth with architecture-level decision quality.
- Optimize Network Automation with Vendor Modules and APIs in a guided scenario using realistic tools, constraints, and quality gates.
- Architect trade-offs, risks, and decision points for Network Automation with Vendor Modules and APIs, then record rationale for stakeholder review.
- Defend a portfolio-ready pipeline reliability report for Network Automation with Vendor Modules and APIs with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 17

6 hours/week + labs

Windows Automation with WinRM and PowerShell Patterns

LEARNING OUTCOMES

- Design the principles of Windows Automation with WinRM and PowerShell Patterns and link them to course outcomes at advanced depth with architecture-level decision quality.
- Optimize Windows Automation with WinRM and PowerShell Patterns in a guided scenario using realistic tools, constraints, and quality gates.
- Architect trade-offs, risks, and decision points for Windows Automation with WinRM and PowerShell Patterns, then record rationale for stakeholder review.
- Defend a portfolio-ready pipeline reliability report for Windows Automation with WinRM and PowerShell Patterns with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 18

6 hours/week + labs

Container and Kubernetes Automation Use Cases

LEARNING OUTCOMES

- Design the principles of Container and Kubernetes Automation Use Cases and link them to course outcomes at advanced depth with architecture-level decision quality.
- Optimize Container and Kubernetes Automation Use Cases in a guided scenario using realistic tools, constraints, and quality gates.
- Architect trade-offs, risks, and decision points for Container and Kubernetes Automation Use Cases, then record rationale for stakeholder review.
- Defend a portfolio-ready pipeline reliability report for Container and Kubernetes Automation Use Cases with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 19

6 hours/week + labs

Operational Reporting, Logging, and Governance Standards

LEARNING OUTCOMES

- Design the principles of Operational Reporting, Logging, and Governance Standards and link them to course outcomes at advanced depth with architecture-level decision quality.
- Optimize Operational Reporting, Logging, and Governance Standards in a guided scenario using realistic tools, constraints, and quality gates.
- Architect trade-offs, risks, and decision points for Operational Reporting, Logging, and Governance Standards, then record rationale for stakeholder review.
- Defend a portfolio-ready pipeline reliability report for Operational Reporting, Logging, and Governance Standards with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
- Publish the week output into your cumulative portfolio with concise outcome narrative and proof artifacts.

WEEK 20

6 hours/week + labs

Capstone: Enterprise Automation Pipeline and Operations Handover

LEARNING OUTCOMES

- Design the principles of Capstone: Enterprise Automation Pipeline and Operations Handover and link them to course outcomes at advanced depth with architecture-level decision quality.
- Optimize Capstone: Enterprise Automation Pipeline and Operations Handover in a guided scenario using realistic tools, constraints, and quality gates.
- Architect trade-offs, risks, and decision points for Capstone: Enterprise Automation Pipeline and Operations Handover, then record rationale for stakeholder review.
- Defend a portfolio-ready pipeline reliability report for Capstone: Enterprise Automation Pipeline and Operations Handover with measurable success criteria and next actions.
- Track measurable progress using rubric scores, defect/risk trends, and evidence completeness each week.
- Run a short retrospective focused on what to retain, improve, and scale into the following week.
- Incorporate peer or mentor feedback and revise the week deliverable to professional publication quality.
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Capstone Projects

Project 1: Reusable Role and Playbook Framework

Develop a modular automation framework with tested roles and environment-aware variables.

- Role library with linting, test runs, and idempotency verification
- Inventory and variable hierarchy design with documentation
- Decision summary for structure, maintainability, and reuse patterns

Project 2: Secure Automation Control Plane

Implement enterprise controls for credentials, approvals, execution visibility, and change safety.

- RBAC and secret management integration with workflow templates
- Audit-ready execution reporting and operational health checks
- Risk and mitigation log for production rollout scenarios

Project 3: Enterprise Automation Handover Package

Deliver a complete automation operations package for team handover and governance review.

- End-to-end pipeline demonstration with validation evidence
- Runbooks, escalation paths, and rollback procedures
- 90-day roadmap for scale, reliability, and policy maturity

This curriculum is an official Hexadigitall learning plan. Module sequencing and project details may be refined as industry requirements evolve.